Customer Shopping Behavior Analysis

A complete data analytics workflow uncovering patterns, trends, and business insights from customer purchase data.

Overview

This project analyzes customer shopping behavior to identify patterns, trends, and useful business insights. The complete data analytics workflow includes data loading, Exploratory Data Analysis (EDA), data cleaning, SQL analysis using PostgreSQL, Power BI dashboard development, report creation, and project presentation.

The goal of this project is to demonstrate practical data analytics skills using **Python**, **SQL**, and **Power BI**.

Dataset

age	category
28	Apparel
41	Footwear
36	Accessories
24	Apparel
52	Outerwear
31	Accessories
45	Apparel

The dataset contains customer shopping and purchase-related information used to analyze customer behavior and sales patterns.

Dataset statistics

- 3,000+ total records
- 9 core features
- Structured customer-level transaction data

Data quality notes

- Numeric fields such as age and purchase_amount support direct trend analysis
- Categorical fields like gender, category, season, location, and shipping_type are ready for segmentation
- Suitable for checking completeness, consistency, and purchase pattern distributions before modeling

Key areas analyzed include:

- Customer demographics such as age, gender, and location
- Gender-wise purchasing behavior across item_purchased and category
- Purchase amounts and spending patterns by customer segment
- Product categories, items purchased, and seasonal demand shifts
- Shipping_type preferences and their relationship to purchase behavior
- Sales and revenue insights based on purchase_amount trends

Tools & Technologies

Python

Data loading, cleaning, and Exploratory Data Analysis

PostgreSQL & pgAdmin

SQL queries and database management

Power BI

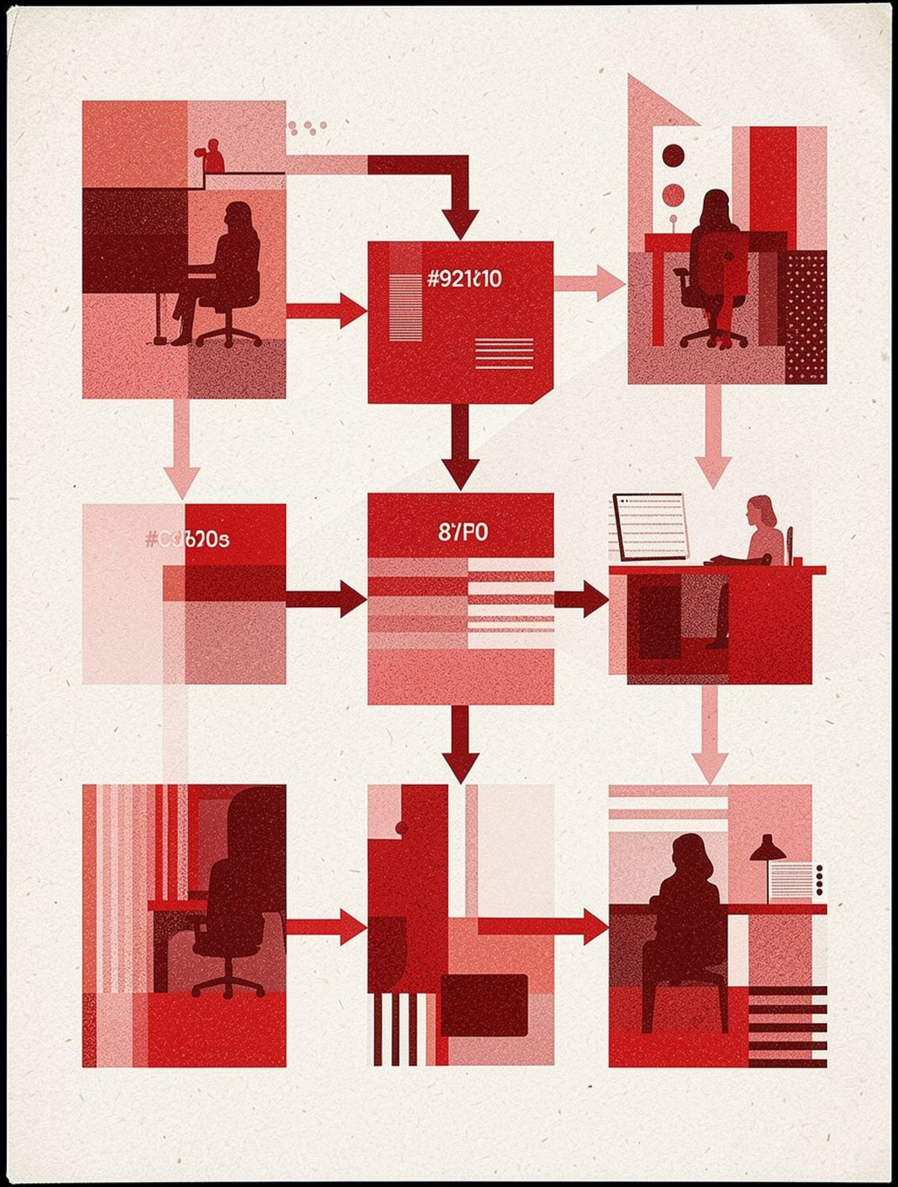
Interactive dashboard creation and visualization

Jupyter Notebook

Python analysis and documentation

Pandas, NumPy & Matplotlib

Data manipulation and visualization



Project Workflow

A structured, end-to-end analytics process from raw data to actionable insights.

Steps 1 & 2 – Data Loading & Exploratory Analysis

1. Data Loading

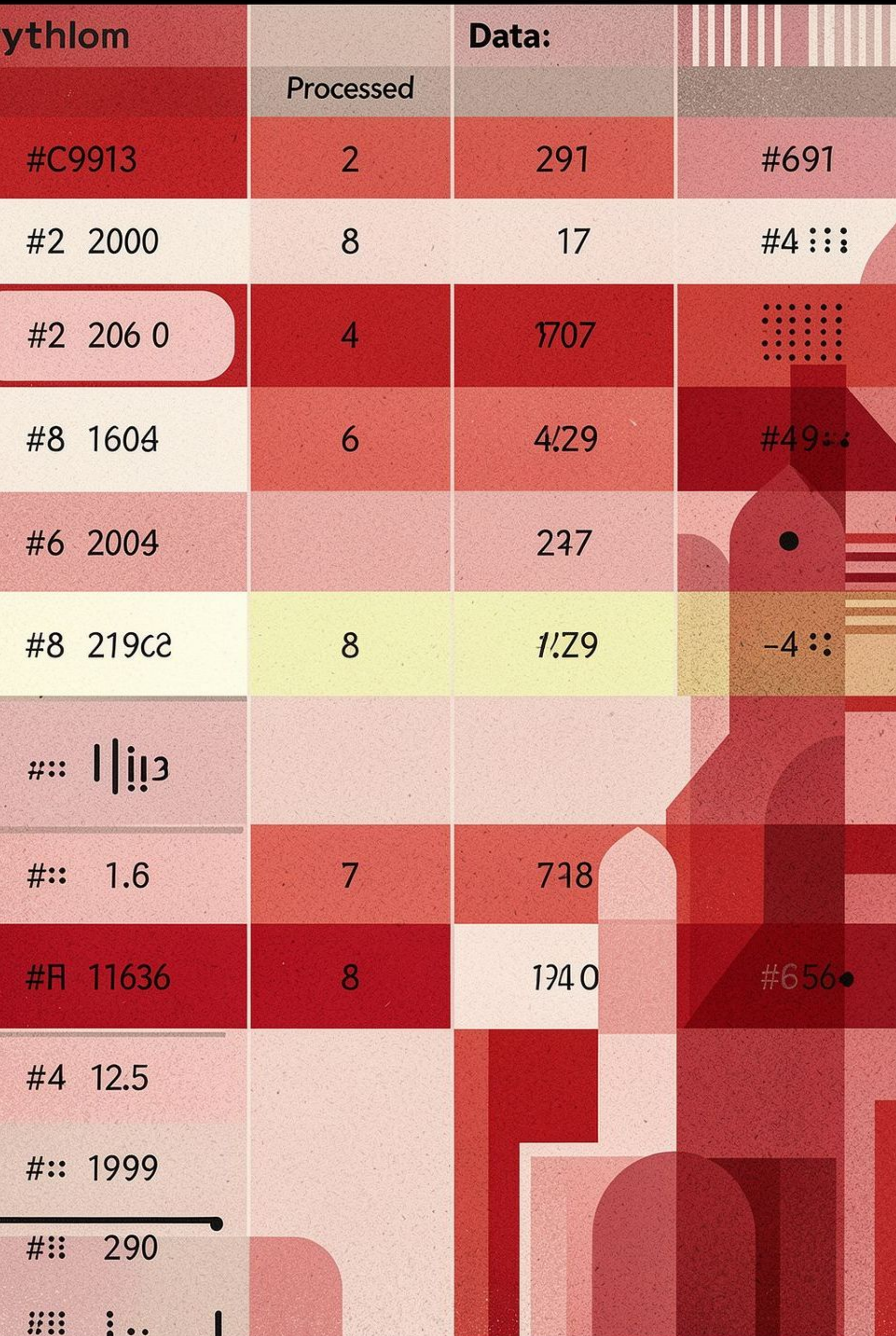
The dataset was loaded into Python using **Pandas** for initial exploration and analysis.

This is the foundation of the entire workflow — ensuring data is accessible and ready for investigation.

2. Exploratory Data Analysis (EDA)

EDA was performed to understand the structure and characteristics of the dataset. The analysis included:

- Checking dataset shape and data types
- Identifying missing values
- Reviewing summary statistics
- Exploring customer and purchase behavior
- Analyzing important patterns and trends



Step 3 – Data Cleaning

The dataset was cleaned and prepared for analysis. Steps included:

01

Handle Missing Values

Checking and resolving any null or incomplete records in the dataset.

02

Remove Duplicates

Identifying and eliminating duplicate records to ensure data integrity.

03

Standardize Columns

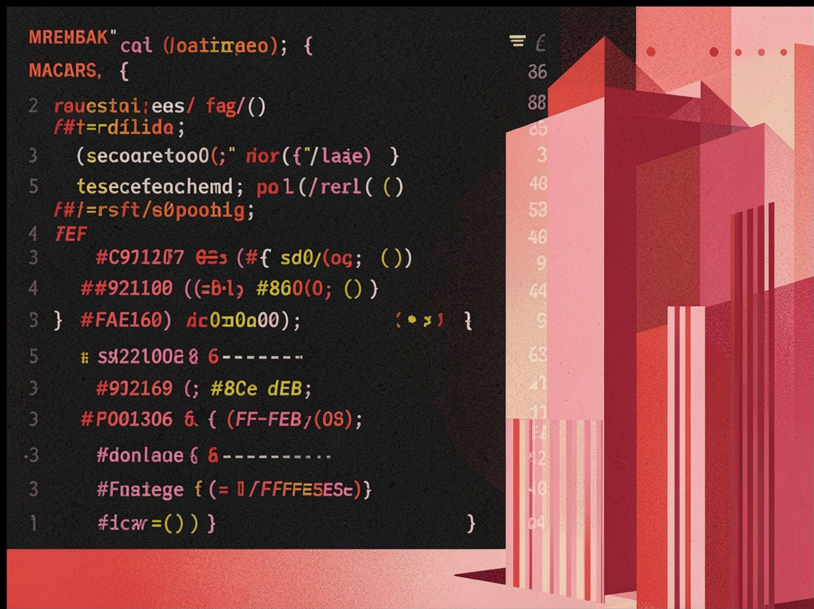
Standardizing column names and correcting data types where required.

04

Prepare for Analysis

Exporting clean data ready for SQL and Power BI analysis.

Step 4 – SQL Analysis Using PostgreSQL



The cleaned dataset was loaded into a **PostgreSQL** server and analyzed using SQL queries. SQL analysis included:

- Revenue analysis by customer gender
- Customer spending analysis
- Discount-related analysis
- Product and category analysis
- Identifying customers with higher purchase amounts
- Using aggregate functions such as `SUM()`, `AVG()`, and `COUNT()`
- Using `GROUP BY`, filtering, and window functions where required

SQL Analysis Results

item_purchased	Average Product rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Step 5 – Power BI Dashboard

An interactive Power BI dashboard was created to present the analysis visually. The dashboard helps users understand:

KPI Cards

Total revenue, average purchase, customer count, and top category at a glance.

Purchase by Category

A bar chart highlights which categories drive the highest purchase amounts.

Gender Distribution

A pie chart shows how purchases are distributed across customer genders.

Seasonal Spending Trends

A line chart tracks spending patterns over time to reveal seasonal shifts.

Step 6 – Project Report

A detailed project report was prepared to document every phase of the analytics workflow — from objective to final findings.

Report sections include:

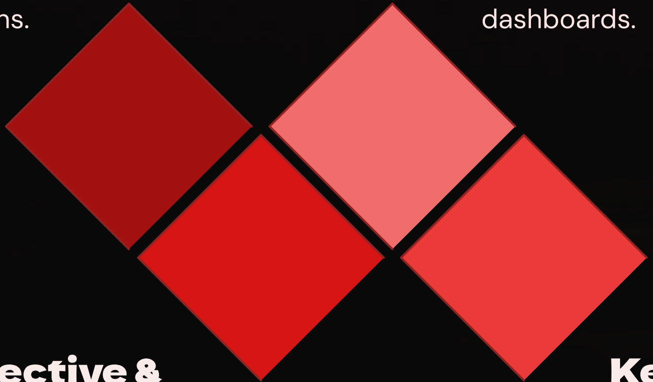
- Project objective
- Dataset information
- Tools and technologies used
- Data cleaning process
- Python EDA
- SQL queries and analysis
- Power BI dashboard
- Key findings and results

Data Cleaning & EDA

Clean data and explore distributions.

SQL Analysis & Power BI

Query data and build dashboards.



Objective & Dataset

Define goals and describe dataset sources.

Key Findings & Report

Summarize insights and recommendations.

End-to-End Analytics – Delivered

This project demonstrates a complete, professional-grade data analytics workflow — from raw data to interactive dashboards and documented insights.

Python & EDA

Data loading, cleaning, and exploratory analysis

SQL

Deep-dive analysis with PostgreSQL queries

Power BI

Interactive dashboards for business stakeholders

Report

Fully documented findings and methodology